Explainability and Trustworthiness in Retrieval Augmented Generation (RAG) Systems - A review of the literature

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Abstract*— Large Language Models (LLMs) are increasingly being integrated into information retrieval and decision support systems; there has been a continued challenge of relying on the fluency of the LLM's output without being factually accurate. Retrieval-Augmented Generation (RAG) has been identified as a leading methodology for improving grounding via the inclusion of external knowledge in inference. However, recent investigations indicate that relying solely on retrieval does not provide satisfactory output because output quality can still be affected by the correctness or incorrectness of reasoning, ambiguous context, and/or an adversary's influence over the output. Hence, this literature review will examine the shifting focus from traditional accurate evaluation measures to include other criteria for evaluating LLM outputs including faithfulness, interpretability and user trust in LLM-generated outputs. Furthermore, the review synthesizes findings from frameworks, explanation and governance studies to contend that developing trustworthy methods of employing RAG must include methods for evaluating retrieval effectiveness in addition to transparency and trust-based evaluation procedures.

Keywords—Retrieval Augmented Generation (RAG), Artificial Intelligence (AI), Large Language Models (LLMs), Explainable AI (XAI), Trustworthy AI (TAI), Human-Centered AI

I. Introduction

Large Language Models (LLMs) are revolutionizing the field of natural language processing by getting high quality, fluent text generation for many applications including decision making, retrieving information, and conversing with humans, in many different contexts[18]. However, LLMs have a well-known problem: since they use probabilistic language modelling to estimate and create text, they often produce inaccurate or fictional text output. Recently, the Retrieval-Augmented Generation (RAG) paradigm has emerged as a way to link model results to external sources of information or knowledge (called "grounded" results)[18][23]. RAG combines retrieval mechanisms with generative architectures so that systems can access current and domain relevant information without having to retrain the parameters of the LLM being used.

Recent research papers have shown that while RAG may help factual consistency in many cases, retrieval alone does not ensure a reliable output[18]. For example, benchmarks such as the CRAG suggest that retrieval provides limited gains in performance[18].

Additionally, datasets like RAGTruth show that when documents related to the generation of an output document are retrieved, it is still possible for hallucination to occur[11]. Therefore, the challenges of reliability of RAG systems come from reasons other than gaps in knowledge and may arise from reasoning errors, ambiguity of retrieved evidence, and misinterpretation of context during generation[23].

As a result, there is a motivation to explore new methods to systematically assess trust and reliability[2]. This includes the birth of sampling based approaches like SelfCheckGPT which provides an alternative to the use of external databases from which to perform checks for reliability directly at generation time through hallucination detection[21]. Also, there are evaluation standards such as TruthfulQA which reveal that models will produce many valid-sounding responses that are incorrect [22].This indicates that models should also be evaluated on truthfulness metrics, not just the traditional metrics of accuracy[18].

Architectural advances aim to incorporate reliability into generation processes in addition to detection. The Self-RAG expands traditional RAG processes by integrating critiques that allow a model to run through the full RAG process (i.e. retrieving, generating and evaluating.)[23]. In addition, related research on faithful reasoning highlights a distinction between statements that are rooted in a retrieved context vs. statements that are objectively true, meaning that there is a need to delineate contextual faithfulness from factual truth[24].

The role of explainability in trusted AI goes beyond evaluating and reasoning; it also serves as a key support for "trust" for AI systems[2]. Recent studies on Explainable AI have established Attribution Methods, Counterfactual Analyses, and Perturbations as the main tools used to provide understanding for model decisions[2]. These methods of providing transparent insight into the data that is retrieved to produce generated outputs allow for robustness analysis of generated outputs based on differences in the way retrieved data[2].

Furthermore, research indicates the impact of trustworthy and ethical AI on the reliability of AI systems as not simply being technical in nature but

rather also socio-technical; that is, they require the inclusion of governance frameworks as well as transparency to build user trust[2]. Diagnostic toolkits and human-centered evaluations both support the development of systems designed for interpretability and for calibrated trust in the AI systems[24].

The evolving literature suggests that it is necessary to perceive RAG as a completely reliable resource[23]. Deployable trust requires the combination of both retrieval quality with both evaluation metrics, hallucinations detection, fidelity of reasoning and explainability frameworks[2]. Therefore, this review will refer to the developing body of work addressing such dimensions in order to show the need for trustworthy layers in modern RAG systems[23].

II. BACKGROUND

LLM

Large Language Models (LLMS) are profound learning systems designed from vast amounts of text to produce human language like results by using a transformer-based structure to help the model learn how to associate contextual ties between words while generating sensible and cohesive responses. The deep linguistic capability that is inherent in any large language model often results in an inability for the large language model to produce the desired result when a great amount of hallucinated results are produced.where in a LLM cannot provide truthful results due to relying on probabilistic forms of language behaviour rather than relying on deduced logical reasoning.

RAG

The Hybrid model of RAG exists as a way forward should a LLMS provide improved performance by providing additional knowledge-based sources to assist in the generating of replies via LLM systems. In an RAG-based system, there exists the ability of providing a knowledge based source(s) and matching that logic against retrieved document(s) that can potentially be used in the context that can improve the LLM-generated response(s). By providing additional reality-based source(s) as being prior or upon completion of retrieval will not only make it possible for newly generated documents with LLMs to have factual consistencies in their nature (e.g., will be dependent upon the continued enhancement of the LLM with additional qualified knowledge obtained from retrieval). However, because of the enhanced quality of knowledge via retrieval of RAG-based sources, the LLM produced from any information included as a part of its model will produce improved results.

XAI

Explainable AI (XAI) is concerned with the interpretability and transparency of decisions made by models.The primary techniques used in these contexts are post-hoc explainability methods, including:

- **LIME (Local Interpretable Model-Agnostic Explanations):** LIME generates local explanations by using an interpretable surrogate model to estimate the predictions of a complex model in its neighbourhood [3]. For text based models, this means determining which words or phrases have had the greatest impact on the model's output [6]. This provides a form of local interpretability for all types of black-box behaviour regardless of whether they are from traditional ML models or LLM's [6].
- **SHAP (Shapley Additive Explanations):** SHAP assigns how important each of the model's inputs is to making a prediction for a given example by computing their contribution score and providing a visual representation of the attribution of the inputs to the prediction made by the model[3]. As a result, SHAP provides greater transparency of the decision-making process and accountability for the models it produces [3].

Common techniques used to investigate how input information impacts the output of generative systems through XAI are attribution analysis, perturbation testing, and counterfactual reasoning.These methods are used to determine which input(s)/retrieved documents contribute towards the generated results and to aid in understanding how grounding information is used in RAG systems during generation.

Hallucination detection includes methods used to determine whether generated outputs have insufficient or contradictory evidence to support the output. Hallucination detection methods evaluate if generated responses are consistent with their respective retrieved context and/or if they are consistent with known facts. Recent approaches allow detection to occur without needing to rely on external verification databases, but instead utilize the consistency/uncertainty of responses

TAI

Trustworthy AI includes methods and principles that are directly related to ensuring reliability, transparency, and accountability of AI systems. When evaluating the trustworthiness of LLMs and RAG systems, it must be determined how generated responses are grounded, interpretable, and consistent with what users expect.

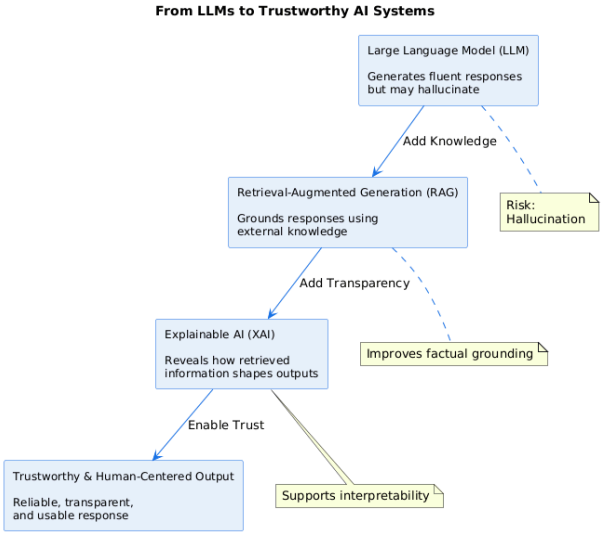


Figure 1: From LLMs to Trustworthy and Human-Centered AI Systems

Figure 1 illustrates the progression from LLMs to more trustworthy AI systems. While LLMs are capable of generating fluent responses, they are prone to hallucinations. RAG enhances reliability by grounding outputs in external knowledge sources. XAI. further improves transparency by revealing how retrieved information influences model responses. Together, these components contribute to the development of trustworthy and human-centered outputs that support reliable and interpretable decision-making.

III. LITERATURE REVIEW

A. ***RAG Reduces Hallucination but Still Requires Trust & Grounding***

RAG (Retrieval Augmented Generation), has become the primary architectural solution to the limitations of LLMs due to their propensity towards generating fluent, yet factually incorrect, hallucinations [11]. This new paradigm of creating answers with external retrieved evidence moves away from relying on parametric memory to relying upon known verifiable data [11]. This change has improved the factually consistent nature of RAG outputs; however, studies have shown that it introduces new forms of reliability issues due to how the retrieved data is interpreted when generating an answer[11]. The reliability issues associated with generated outputs occur as a result of performance differences at an empirical level as well as by various structural ways (mechanisms) in which grounding fails[11].

Empirical Performance Gaps in RAG Systems

Research evaluation has shown that progress has been made with regard to the reliability of RAG; however, empirical evaluation of currently implemented RAG has shown that RAG currently only presents very limited, practical potential improvements of reliability[18]. The CRAG (Comprehensive RAG Benchmark) testing allows us to evaluate RAG in a variety of dynamic real-world knowledge situations and that retrieval-assisted LLMs provide access to improved accuracy when compared to independent LLMs[18]. Nevertheless, even with the benefits of retrieval, the overall performance of RAG remains well below that which can be considered trustworthy on long-tail entities and rapidly changing information[18]. This indicates that retrieval does improve the level of access but does not provide a guarantee for the correctness of reasoning or reasoning accuracy[18].

Through an expansive annotated dataset showing hallucination analysis in grounded contexts has been established[11]. The evidence demonstrates that hallucinations still occur after relevant documents have been retrieved, supporting the notion that most of the errors occur with the model producing claims that expand beyond the context supplied[11]. Although RAGTruth does provide insights into grounding errors, its emphasis is mainly on the detection of these errors instead of on preventing or avoiding them[11].

Mechanisms of Grounding Failure

Recent analysis goes beyond simply analyzing performance metrics by inquiring into situations where even after relevant information has been found, grounding is still unable to occur[12]. In the case of a study titled "LLMs Can Be Fooled into Labelling a Document as Relevant", the authors found evidence that demonstrates how models tend to base their determination of whether evidence retrieved is relevant or not on lexical similarity rather than on semantic similarity[12]. The authors did this using adversarial testing and demonstrated that irrelevant documents containing query-specific terms or keywords were incorrectly treated as valid sources upon which the model could ground its response, exposing a vulnerability in the retrieval phase of the pipeline[12].

This broad systemic vulnerability is elaborated upon in

[13] where the authors framed hallucination as an issue at the pipeline level, which results from the interaction of retrieval quality, generation characteristics, and verification processes. The authors indicated that treating hallucination as a "single-point failure" is less important than understanding that a model's reliability is based on all of the stages of a Robust Architecture Generation (RAG) pipeline working together in a coordinated fashion[13].

Emerging Need for Trust-Oriented Mechanisms

Overall, these findings indicate that RAG transfers the hallucination problem from solely being one of knowledge access to now being one of both reasoning and evaluating information[18]. Retrieval improves the ability to ground retrieved material to reality; however, additional mechanisms must be in place to ensure the reliability of outputs produced by processes using retrieved information to determine how the retrieved material was used during generation[13]. This has created an increasing interest in the development of trust-aware approaches to RAG pipelines[18].

B. **Retrieval Strategy Determines Reliability**

Retrieval-Augmented Generation provides an improvement to factual grounding, but recent studies have demonstrated that factors such as document relevance and the configuration of retrieval effects are also responsible for determining how well RAG produces useful outputs[14]. Traditional approaches usually regard document relevance as being enough in order to achieve useful output through traditional methods; however, recent research shows that retrieval strategy has the most significant impact on both the quality of reasoning and the overall reliability of the system[16]. Many of these same conclusions can be seen in the sensitivity of the pipeline to document types or categories, the degree of ambiguity associated with the retrieved document, and the degree of influence on reasoning through diversity in retrieved documents and the adversarial environment surrounding document retrieval[15].

Sensitivity to Retrieval Configuration

A systematic review of RAG (Retrieval-Augmented Generation) workflow has underscored the significance in how retrieval is configured [14]. The report discusses the various components of a RAG pipeline (such as query classification, rereferencing, and summarizing) and has determined that the performance of the pipeline is highly affected by the composition of the retrieved contextual data, and especially by the amount of retrieved data that solely answers the query being processed [14]. Thus, retrieval effectiveness is dependent upon not only the relevance of the documents retrieved but also upon how the retrieved content is organized and used during the generation process [14].

In a similar understanding, the paper [15] investigated ambiguous instances of multiple documents that appear

equally relevant and identified that ambiguity of semantics may exacerbate hallucination risk by creating channels of difficulty for the model as it tries to evaluate which and/or the most relevant grounding source to utilize. Consequently, the overall reliability of the model is less about the presence of relevant documents and is more contingent on the clarity of the retrieved dataset [15].

Diversity and Noise in Retrieved Context

Recent studies indicate that reasoning performance can be influenced by the presence of returned documents that vary from one another. This idea is explored in the paper [16] which looks at the impact of varying the collection of retrieved documents to include randomly selected (or diverse) documents, on the generations produced by a retrieval-augmented generation (RAG) system [16]. Unexpectedly, this work finds that including random documents can improve generation accuracy for some of the examples evaluated[16]. As such, this implies that a strategy of returning only closely matched documents may limit success when reasoning, while more broadly varied sets of retrieved documents may improve success when reasoning under certain situations [16].

Vulnerability to Adversarial Retrieval

Retrieval methods may not be impervious to adversarial influence as outlined in the work LLMs Can Be Fooled into Labelling a Document as Relevant shows, because LLMs appear to largely rely upon lexical overlap when determining whether or not a document is relevant to a query [12]. Therefore, it is possible for an irrelevant document that contains some of the words contained within the query to be evaluated as valid grounding for the given response to the query [12]. This indicates that retrieval cannot simply be viewed as a passive input stage, but rather a dynamic component that influences the validity of the response that gets generated for the query [12].

Collectively, these research studies suggest that there are some things that affect retrieval reliability: configuration, diversity, and robustness; rather than just relevance[14][12]. Therefore, in order to create trustworthy RAG systems we must pay attention to not just what information is retrieved, but also to how the information is selected and interpreted [14].

C. Explainability in AI Systems and Trustworthiness

The deployment of modern Large Language Models (LLMs) into real-world applications presents an obstacle to many users due to their "black-box" nature that provides little to no explanation for the generated output [6]. Users of the LLM are often not provided with a good enough rationale for the results provided to them to establish a sense of trust, accuracy, and accountability, nor to allow for responsible deployment of Artificial Intelligence [6][4]. Similarly, the lack of transparency when predicting in machine learning for Botnets results in a numerical classification that does not provide a good enough rationale to users of those applications [3].

To address the problems with transparency across various domains, Explainable AI (XAI) methods need to have improvements made using proper explanation principles [3]. Explanations must be created specifically for end-users, respond to the "why" of a situation rather than just the "what" of it, utilize different types of forms and methods when appropriate, as well as be tested appropriately for both effectiveness and reliability [4]. Furthermore, the explanations of how information is presented will also play an important role in trust developing between users and their interactions with AI systems [6].

2. Methodologies and Techniques

In Various domains, different techniques of explainable artificial intelligence (XAI) are applied to binary and multi-class classification problems using/with different types of prediction models such as extreme gradient boosting (XGBoost) on various instances of the MedBIoT, Bashlite, Mirai, and Torii datasets [3] and LLMS. These different classes of prediction techniques achieve high prediction accuracy, however, their lack of interpretability reiterates the need for explainability mechanisms on a more systematic basis [3]. Moreover, there is much more analysis needed on the inner workings of large language models (LLMs) in order to enable the ethical development and/or deployment of AI applications [6].

In contrast, interpretability for large language models (LLMs) creates additional difficulties due to their immense transformer architectures and their extremely high number of parameters [5]. Although post-hoc techniques provide information about tokens or attribute-level features, additional in-depth methodology is needed to understand the ways in which the models internally reason as well as verifying that they align with important AI principles such as responsible AI [5]. To facilitate the incorporation of explainability into LLM pipelines holistically, recent architectural models have proposed embedding an XAI layer between the core model and the output interface utilizing attention visualizations, feature importance analysis, and counterfactual explanation techniques to yield structured insights into model reasoning. In addition to this, these models are using iterative evaluation processes (in order to assess interpretability, fidelity, and coherence) to continually refine both the predictive power and the transparency of their LLMs relative to responsible and trustworthy AI principles [5].

D. Emerging Methodologies for RAG Systems

While general interpretability techniques for LLMs are effective, specifically for Retrieval Augmented Generation (RAG) systems to allow the user to validate where the information originated from are important to have. LLMs often make unsupported claims, as well as factual inaccuracies and RAG is designed to provide additional context about the model by providing additional facts that were retrieved from information that is relevant to the question; but both systems can still generate contradictory statements with respect to what was retrieved from the information source[11].

To address this, several specialized methodologies have emerged like Faithful Reasoning and Provenance Tracing: A critical methodology for building trust is ensuring that a system's underlying computations mirror logical reasoning. By forcing a model to explicitly select statements from a context and draw inferences solely from that selection, the system can expose its reasoning trace to the user. This prevents the model from relying on hidden, potentially biased internal weights, allowing users to verify how the final answer was reached[19].

Attribution Evaluation: Trustworthiness in RAG relies on the changing quality of attribution from one step to another, determining how closely related the output generation to the source data supporting it. There are several methodologies for evaluating how closely support matches the retrieved document, and these include evaluating the relationship between spans of text, relative to the totality of the supporting evidence, that can be classified as providing full support, partial support, or creating a contradiction between the supporting evidence and the final generated output [20]. This methodology relies on identifying discrepancies in the relationship between the retrieved information to generate a response [20].

Self-Reflection and Critique: Recent frameworks with the emergence of newer methodologies, like SELF-RAG, during the inference phase, the model is trained in an effort to produce "reflection tokens" as the inference is taking place. These self-reflection tokens are designed to allow the model to evaluate whether the documents it retrieved are needed and whether they are relevant and if the output generated by the model is fully supported by the evidence used for output generation [20].

Informational Consistency and Perturbation: to identify hallucinations and assess the effect of inputs without the need for external sources of information. This is based on the idea that if a model has factual knowledge, the responses generated from random sampling would remain stable, but if the model is generating fictitious responses, these responses will differ from one another and provide contradictory information. The concept of observing output variation due to changes in inputs is similar to the methods used in perturbation-based attribution. [21]

E. Human Centered Trustworthiness

Recent studies on Human-Centered AI indicate that as these kinds of systems become common place, there needs to be less emphasis on purely technical performance and more of an emphasis on holistically understanding how trust in AI works, how trust in AI can be explained to humans, and how a systematic view can be applied in creating reliable AI systems [1].

Trust between humans and machines has emerged as a key consideration when interacting with AI, but there is currently no universally accepted framework for the evaluation of trustworthiness of different types of AI systems (i.e. robots, smart vehicles, decision support tools) making it difficult to determine the degree or level of trust that society places in them [9]. Difficulties in the assessment and measurement of trust are compounded by the move from deterministic computer programs to self learning algorithms where an individual user is unable to completely understand how the AI system has come to its independent decision-making capabilities [2].

Reviewing them also identifies a paradox in which there is typically an inverse correlation between the accuracy of a model's predictions and its interpretability. Although highly accurate complex models (e.g., deep neural networks) tend to achieve the best performance, they are typically least interpretable and are often referred to as "black boxes." This lack of interpretability can create challenges in critical domains such as healthcare and autonomous driving. In order to promote transparency in understanding complex black box models, researchers distinguish between two types of transparencies: (1) transparent design, where models have a built-in capability for transparency, and (2) post-hoc explanation, whereby researchers use surrogate models to provide approximations of complex models and explain the model behavior[2].

In addition to the notion of technical transparency, a human-centered approach to explainable AI (XAI) aims to provide practically useful information that enable effective collaboration between the user and the system[4]. Empirical findings have indicated that the majority of end-users do not require a complete representation of the underlying technical details of a model; rather, they prefer to receive part-based explanations of the model that are consistent with their own cognitive processes (e.g., breaking an image down into individual concepts)[4]. The paradigm of "Help Me Help the AI" promotes the notion that end-users can use the explanations provided by the model to enhance their understanding of the model, develop their own skills in performing tasks associated with the model, and change their behaviors to provide better inputs to the model[4].Furthermore, research highlights that trust is a multidimensional construct, and XAI can be effectively processed through affective means by non-technical users[1].

Finally, the part of this transition to TAI(Trustworthy AI) will be to create an overall model that includes integrated frameworks for navigating threats across the AI lifecycle, such as the SAIT Nav matrix (Strategic AI Threats Navigation)[7]. These integrated frameworks are necessary to mitigate the inherent risks associated with generative models (hallucinations), data poisoning, and the potential of biases being inherently built into algorithms. The importance of human-sentient design is critical in high-consequence environments, such as autonomous transportation, where there must be a high degree of concern for the safety of humans when employing systems that are highly automated and have a pre-existing social context[7]. By connecting the fragmented areas of data protection, robustness, and accountability, the goal is to ensure that AI remains a tool for enhancing human values and improving the quality of the planet, rather than threatening either[7].

3. Evaluation of Trustworthy ai

To put into practice the four FEAS categories (Fairness, Explainability, Auditability, and Safety) and build system trust requires a comprehensive evaluation framework. Trustworthy AI (TAI) incorporates ethics consistent with human values, applicable to how those value systems affect AI decisions and behaviours. It is based principally upon self assessment as well as TAI evaluation and its goal is to develop and use AI in an ethical manner with respect to safety[10][8].TAI's evaluations characterises evaluation methods as belonging to one of the four categories according to their maturity and the type of solution they employ; these categories are conceptual, manual, automated, and semi-automated evaluation methods[10]. Typically, evaluation methods will be divided into sub-categories including but not limited to the following: fairness and compliance evaluations; transparency

evaluations; risk and accountability evaluations; and trust and safety evaluations.

- Conceptual Evaluation Methods: highlevel frameworks that deal with multiple aspects of trustworthy AI throughout its lifecycle but do not have implementation details or testing or validation.
- Manual Evaluation Methods: structured questionnaire developed to EU TAI values. Provide an in-depth assessment; however, there are time constraints, and insufficient external data is available for the assessment[10].
- Automatic Evaluation Methods: technical metric-based tools, allowing for uniform and scalable evaluation of existing traits of trustworthy AI; however, evaluation characteristics may be limited to pre-defined metrics, and there may be challenges in capturing fairness concerns in real world situations.
- Semi automated Evaluation: approaches which involve both automation and human judgement, provides flexibility, and greater preference for evaluation within context-sensitive environments (healthcare) where contextually relevant knowledge is required for fairness assessment.

IV. CONCLUSION

The introduction of LLMs into information retrieval systems has directed interest away from producing coherent text to developing factual accuracy, with RAG being the primary answer to hallucinations. This review demonstrates that RAG is an excellent way to improve grounding by providing access to external information, but has issues related to reliability, as systems that rely solely on retrieval are still susceptible to reasoning errors, ambiguous context, and adversarial attacks. Reliability in literature is determined by the effective organization of the retrieval pipeline and the model's capability to give a faithful representation of complex or noisy data.

Therefore, conventional measures of accuracy are now inadequate and new evaluation frameworks that encompass the entire life cycle of generating end results will need to be established. Future development will need to focus on explainable AI (XAI) and trustworthy AI (TAI) principles on RAG to close the gap between capabilities and reliability. Attribution analysis, self-reflection, and perturbation testing are some techniques that can facilitate the conversion of "black box" results to transparent, verifiable, and reputable reasoning processes.

Building on these findings, our project aims to address the identified gap between technical reliability and user-understandable transparency in RAG systems. While existing studies largely focus on improving retrieval performance or detecting hallucinations, there remains limited work on making the reasoning process of RAG outputs interpretable in a practical and human-centred way. To address this, our research will develop a lightweight RAG prototype incorporating document influence scoring and perturbation-based attribution to reveal how retrieved evidence shapes generated responses. By evaluating these mechanisms using both automated reliability metrics and human-centred feedback, the project seeks to explore whether simple explanation techniques can enhance user trust, clarity, and the ability to assess output reliability.

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